

QUESTIONS 0 / 19 completed

Plus

PLUS

Write a C++ program to overload the + operator to add two complex numbers.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Complex {
5     int real, imag;
6
7 public:
8     Complex(int r = 0, int i = 0) {
9         real = r;
10        imag = i;
11    }
12
13    Complex operator+(Complex c) {
14        return Complex(real + c.real, imag + c.imag);
15    }
16
17    void display() {
18        cout << real << " + " << imag << "i";
19    }
20 };
21
```

INPUT

```
3 4
5 6
```

```
17 void display() {  
18     cout << real << " + " << imag << "i";  
19 }  
20 };  
21
```

OUTPUT

```
8 + 10i
```



QUESTIONS 2 / 19 completed

Increment



INCREMENT

Write a C++ program to overload the ++ operator to increment a variable.

PROGRAM EDITOR

C++



✓ Saved

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Number {
5     int x;
6
7 public:
8     void input() {
9         cin >> x;
10    }
11
12    void operator++() {
13        ++x;
14    }
15
16    void display() {
17        cout << x;
18    }
19 };
20
```

INPUT

10

```
20
21 int main() {
22     Number n;
23
24     n.input();
25     ++n;
26     n.display();
27
28     return 0;
29 }
```

OUTPUT

```
11
```

QUESTIONS 4 / 19 completed

<< ▾

<<

Write a C++ program to overload the << operator to print the contents of a user-defined class.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Point {
5 private:
6     int x, y;
7
8 public:
9     Point(int x = 0, int y = 0) : x(x), y(y) {}
10
11     friend ostream& operator<<(ostream& out, const Point& p);
12 };
13
14 ostream& operator<<(ostream& out, const Point& p) {
15     out << "(" << p.x << ", " << p.y << ")";
16     return out;
17 }
18
19 int main() {
20     int a, b;
21     cin >> a >> b;
```

```
17 }  
18  
19 int main() {  
20     int a, b;  
21     cin >> a >> b;  
22  
23     Point p(a, b);  
24     cout << "Point: " << p << endl;  
25  
26     return 0;  
27 }
```

OUTPUT

```
Point: (0, 16)
```

QUESTIONS 4 / 19 completed

Equal

EQUAL

Write a C++ program to overload the == operator to compare two objects of a user-defined class.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Complex {
5 private:
6     int real, imag;
7
8 public:
9     Complex(int r = 0, int i = 0) {
10         real = r;
11         imag = i;
12     }
13
14     void getInput() {
15         cin >> real >> imag;
16     }
17
18     bool operator==(const Complex &obj) {
19         return (real == obj.real && imag == obj.imag);
20     }
21 };
```

```
15     cin >> real >> imag;
16 }
17
18 bool operator==(const Complex &obj) {
19     return (real == obj.real && imag == obj.imag);
20 }
21 };
22
23 int main() {
24     Complex c1, c2;
25     c1.getInput();
26     c2.getInput();
27
28     if (c1 == c2)
29         cout << "Both objects are equal" << endl;
30     else
```

OUTPUT

```
Both objects are equal
```


QUESTIONS 5 / 19 completed

[]



[]

Write a C++ program to overload the [] operator to access the elements of an array using an index.

PROGRAM EDITOR

C++



Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 class Array {
5     int a[100];
6 public:
7     Array(int n = 0) {
8         for(int i = 0; i < n; i++) a[i] = 0;
9     }
10
11     void setValue(int index, int value) {
12         a[index] = value;
13     }
14
15     int& operator[](int index) {
16         return a[index];
17     }
18 };
19
20 int main() {
21     int n;
```

INPUT

```
5
10 20 30 40 50
3
```

Write a C++ program to overload the [] operator to access the elements of an array using an index.

```
23
24     Array arr(n);
25
26     for (int i = 0; i < n; i++) {
27         int x;
28         cin >> x;
29         arr[i] = x;
30     }
31
32     int idx;
33     cin >> idx;
34
35     cout << arr[idx];
36
37     return 0;
38 }
```

OUTPUT

40